From Pythagoras to Transformers: A Geometric Theory of Attention via the Relation Body

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Abstract

Understanding the internal mechanisms of large language models (LLMs) remains a central challenge in artificial intelligence. This paper proposes a unified geometric framework—the Relation Body—which decomposes the relationship between any two vectors into two orthogonal states: the "same/redundant" component measured by the inner product, and the "different/independent" component measured by the outer product. Their normalized forms satisfy $c^2 + s^2 = 1$, i.e., the Pythagorean theorem on the unit circle. We apply this framework to the attention mechanism of Transformers and establish four results: the Relation Body Theorem (Theorem 1, a strict theorem), the Redundancy-Probability Empirical Theorem (Theorem 2), the No-Trunk Theorem (Theorem 3, a strict theorem), and the Redundancy Compensation Empirical Theorem (Theorem 4). Experiments on GPT-2 small and GPT-2 medium reveal that attention exhibits a shrub structure: no global trunk, but local stems.Heads with high first principal component (PC1) concentration correspond to syntactic/positional functions, while heads with low PC1 concentration correspond to semantic functions. Layer 0 and the final layer serve as information entry and semantic exit bottlenecks, respectively, while middle layers possess strong redundancy compensation. Controlled-domain experiments directly verify the core causal mechanism: the narrower the input semantic distribution, the higher the PC1, and the closer the structure is to a traditional tree with a trunk. Based on this, we predict that individual experts in Mixture-of-Experts (MoE) models should exhibit stronger local trunk structures than dense models; this prediction remains to be empirically verified. This work provides a computable, falsifiable geometric theory for understanding how attention organizes information, and opens a new path of "geometric interpretability" for large-model research.

Keywords: attention mechanism, interpretability, relation body, inner and outer product, principal component analysis, mixture of experts

1 Introduction

1.1 Background

Transformer has become the foundational architecture for LLMs (Vaswani et al., 2017), yet its internal mechanisms remain a black box.Existing interpretability methods include attention visualization (Elhage et al., 2021), probing (Tenney et al., 2019), causal tracing (Meng et al., 2022), and sparse autoencoders (Bricken et al., 2023) . However, thereis a lack of a unified geometric theory to explain: why attention is effective? How do multiple heads divide labor? How do layers collaborate?

1.2 Limitations of Existing Methods

Existing methods are bottom-up, local, and isolated. They do not provide a unified explanation for redundancy, inter-layer information flow, or cross-model universality.

1.3 Core Idea: Relation Body

The relationship between any two non-zero vectors can be fully characterized by the inner product and the outer product. Inner product: same/parallel/redundant; outer product: different/orthogonal/independent. After normalization: $c^2 + s^2 = 1$, the unit circle. We view the attention mechanism as an information organization process on the Relation Body.

1.4 Main Contributions

1. We propose Relation Body geometry, prove Theorem 1, and establish Theorem 2, Theorem 3, and Theorem 4. Theorem 1 and Theorem 3 are strict theorems; Theorem 2 and Theorem 4 are empirical theorems supported by experiments, with their theoretical mechanisms left as open problems.
2. We validate the "no-trunk shrub" structure on GPT-2 small/medium.
3. We find that high-PC1 heads are syntactic/positional heads, while low-PC1 heads are semantic heads.
4. We discover a global semantic broadcast head in Layer 11, analogous to apical dominance.
5. Ablation experiments demonstrate entry/exit bottlenecks and middle-layer redundancy; cumulative ablation reveals a phase transition point $k^* \approx 4$.
6. Controlled-domain experiments verify that "narrower semantics → higher PC1 → closer to a tree with a trunk."
7. We predict that individual MoE experts have significantly higher PC1 than dense models; this prediction remains to be verified.

Scope and series context: This paper focuses on establishing the foundational theory and preliminary validation on the GPT-2 model family. A companion paper will address cross-model generalization (e.g., LLaMA, Qwen) and alignment paradigms.

1.4.1 Paper Structure

Section 2 presents the Relation Body geometry and four results. Section 3 describes the experimental setup and metrics. Section 4 reports the main experimental results. Section 5 discusses the findings and MoE predictions. Section 6 concludes the paper and outlines future work.

2 Relation Body Geometry (Theory)

This section establishes the core theoretical framework of the paper—Relation Body geometry. Starting from the fundamental identity of inner and outer products, we prove that the relationship between any two non-zero vectors can be normalized to a point on the unit circle. We then derive four results: the Relation Body Theorem (Theorem 1, a strict theorem), the Redundancy-Probability Empirical Theorem (Theorem 2), the No-Trunk Theorem (Theorem 3, a strict theorem), and the Redundancy Compensation Empirical Theorem (Theorem 4). These results unify the probability distribution in attention, the geometric structure of Value vectors, and inter-layer information flow into a computable and falsifiable mathematical framework.

2.1 Inner Product, Outer Product, and the Unit Circle

2.1.1 Definition of Relation Coordinates

Let $a, b \in \mathbb{R}^d$ be non-zero vectors, $\langle \cdot, \cdot \rangle$ be the Euclidean inner product, and $\wedge$ be the outer product (wedge product). Define the **relation coordinates**:

$$c(a, b) = \frac{\langle a, b \rangle}{\|a\| \|b\|}$$

$$s(a,b) = \frac{\|a \wedge b\|}{\|a\|\|b\|}$$

where the outer product norm is given by:

$$\|a \wedge b\|^2 = \|a\|^2\|b\|^2 - \langle a, b\rangle^2$$

Here c measures the degree of sameness/parallelism between the two vectors, and s measures the degree of difference/orthogonality. They correspond to the normalized forms of the inner and outer products, respectively.

2.1.2 Relation Body Theorem (Theorem 1)

Theorem 1 (Relation Body Theorem)
For any non-zero vectors $a, b \in \mathbb{R}^d$, we have

$$c^2 + s^2 = 1$$

That is, (c, s) lies on the unit circle S^1.

Proof
From the outer product norm identity:

$$\|a \wedge b\|^2 = \|a\|^2\|b\|^2 - \langle a, b\rangle^2$$

Dividing both sides by $\|a\|^2\|b\|^2$:

$$\frac{\|a \wedge b\|^2}{\|a\|^2\|b\|^2} = 1 - \frac{\langle a, b\rangle^2}{\|a\|^2\|b\|^2}$$

That is,

$$s^2 = 1 - c^2$$

Therefore,

$$c^2 + s^2 = 1$$

■

2.1.3 Geometric Interpretation and Information-Theoretic Meaning

Let the angle between a and b be θ. Then

$$c = \cos\theta, \qquad s = |\sin\theta|$$

Thus $(c, s) = (\cos\theta, |\sin\theta|)$, which lies exactly on the unit circle. The angle θ corresponds to the angle on the Relation Body.

Figure 1 shows the schematic of the Relation Body on the unit circle. The horizontal axis is c (same/parallel), and the vertical axis is s (different/orthogonal). The circle marks three extreme relations: parallel ($c = 1, s = 0$), orthogonal ($c = 0, s = 1$), and anti-parallel ($c = -1, s = 0$).

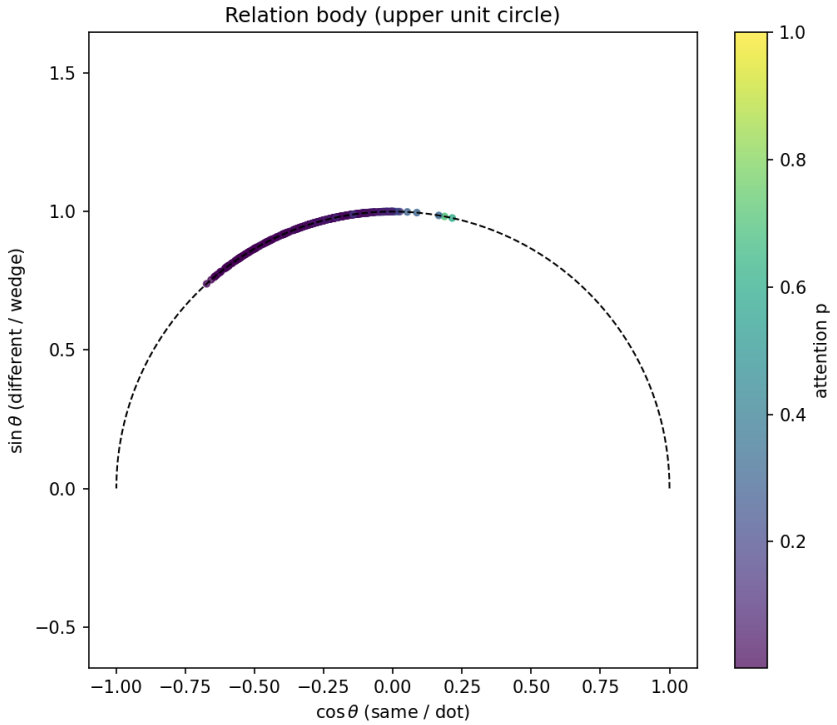


Figure 1: Schematic of the Relation Body on the unit circle. The horizontal axis is c (same/parallel), and the vertical axis is s (different/orthogonal). The circle marks three extreme relations: parallel ($c = 1, s = 0$), orthogonal ($c = 0, s = 1$), and anti-parallel ($c = -1, s = 0$).

Corollary 1.1 (Information Conservation)
The equation $c^2 + s^2 = 1$ can be interpreted as: the proportion of shared information c^2 and the proportion of independent information s^2 between two vectors always sum to 1.

Corollary 1.2 (Connection to the Pythagorean Theorem)
The unit circle equation on the Relation Body, $c^2 + s^2 = 1$, is formally identical to the unit circle $x^2 + y^2 = 1$. Therefore, Relation Body geometry is essentially a natural generalization of the Pythagorean theorem in vector relation space.

2.2 Redundancy-Probability Empirical Theorem (Theorem 2)

2.2.1 Basic Setup of Attention Mechanism

Consider single-head attention in a Transformer. Given an input sequence $X = [x_1, \ldots, x_n] \in \mathbb{R}^{n \times d}$, linear projections yield Query, Key, and Value(QKV):

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_h}$. The attention probability is

$$p_{ij} = \text{softmax}\left(\frac{q_i \cdot k_j}{\sqrt{d_h}}\right)$$

The context-weighted average is

$$\bar{V}_i = \sum_{j=1}^{n} p_{ij} V_j$$

Define the **local difference**:

$$\Delta_{ij} = \|V_j - \bar{V}_i\|$$

A smaller Δ_{ij} indicates that the Value of the j-th token is closer to the context mean of the current query, i.e., higher information redundancy.

2.2.2 Empirical Theorem (Theorem 2)

Theorem 2 (Redundancy-Probability Empirical Theorem). Across all (layer, head) pairs of GPT-2 small and GPT-2 medium, the attention probability p_{ij} and the local difference Δ_{ij} exhibit a stable negative correlation, as shown in Table 1.

Table 1: Redundancy-probability correlation across all (layer, head) pairs.

Model	Mean r_A	Negative correlation ratio
GPT-2 small	-0.6783	100.00%
GPT-2 medium	-0.6657	99.74%

That is:

High attention probability corresponds to high local redundancy.

Remark. This conclusion is supported by experiments (Section 4.1), but its theoretical mechanism remains not fully understood. We attempted to derive it using Assumption A1 ($q_i \cdot k_j \approx \lambda v_i \cdot v_j$), but the empirical verification in Appendix A.5 shows that Assumption A1 does not hold on dense models (mean $r = 0.055$, significant positive correlation ratio only 45.83%). Therefore, Theorem 2 is labeled an **empirical theorem**, and its strict theoretical derivation is left as an open problem.

Relation to Assumption A1. The failure of Assumption A1 is consistent with the "shrub" model—wide semantics lead to no global trunk, which in turn decouples the QK space from the V space. Assumption A1 is expected to hold for individual MoE experts, which constitutes part of the MoE prediction (Prediction 2; see Section 2.5).

2.2.3 Corollaries (Empirical Regularities)

Corollary 2.1 (Negative correlation between entropy and concentration)
Experiments show that attention entropy and local PC1 concentration are negatively correlated (mean $r = -0.41$). This can be understood directly from Theorem 2: low entropy $\rightarrow$ sharp distribution $\rightarrow$ a few tokens dominate $\rightarrow$ their Value vectors are similar $\rightarrow$ high PC1.

Corollary 2.2 (Local weighted variance)
The local difference can be expressed as a weighted variance:

$$\sum_j p_{ij} \Delta_{ij}^2 = \mathrm{Var}_p(V)$$

Corollary 2.3 (Intuitive basis for no global trunk)
If the semantic distribution is wide (high rank), the global PC1 approaches $1/d_h$. This provides an intuitive basis for Theorem 3.

2.3 No-Trunk Theorem (Theorem 3)

From Theorem 2 (an empirical theorem), locally high probability corresponds to high redundancy. We now turn to the global picture: does there exist a "trunk" spanning all tokens across the entire sequence? Theorem 3 will prove that under a wide semantic distribution, the answer is negative. Theorem 3 is a strict theorem, and its derivation does not depend on the experimental results of Assumption A1; the failure of Assumption A1 only affects the strictness of Theorem 2, not the mathematical foundation of Theorem 3.

2.3.1 Assumptions

Assumption A3 (Wide semantics)
The intrinsic dimension of the semantic distribution of the training data, D_{sem}, is much larger than the head dimension d_h:

$$D_{\text{sem}} \gg d_h$$

Assumption A4 (Random projection)
The trained Value projection matrix W_V approximately behaves as a random orthogonal projection in the semantic space, i.e., the rows of V are approximately independent and identically distributed random vectors.The applicability of this assumption will be examined experimentally (see Figure A3 in Appendix C for the SVD spectrum analysis).

2.3.2 Theorem and Proof

Theorem 3 (No-Trunk Theorem)
Under Assumptions A3 and A4, the global PC1 concentration approaches the random baseline:

$$\frac{s_1^2}{\sum_{k=1}^{r} s_k^2} \to \frac{1}{d_h}$$

where $s_1 \geq s_2 \geq \cdots \geq s_r$ are the singular values of the Value matrix V.

Proof
By A4, the rows of V are approximately i.i.d. The eigenvalue distribution of the covariance matrix

$$C = \frac{1}{n} V^T V \in \mathbb{R}^{d_h \times d_h}$$

converges to the Marchenko-Pastur distribution (Marchenko & Pastur, 1967). The MP distribution has no atomic mass, i.e., no isolated large eigenvalues. The largest eigenvalue is approximately

$$\lambda_{\max} \approx \sigma^2 (1 + \sqrt{\gamma})^2$$

where $\gamma = d_h/n$. The sum of all eigenvalues is $d_h \sigma^2$. Therefore,

$$\frac{\lambda_{\max}}{\sum_k \lambda_k} = \frac{(1+\sqrt{\gamma})^2}{d_h} = O\left(\frac{1}{d_h}\right)$$

When d_h is finite, the PC1 concentration approaches $1/d_h$. ■

2.3.3 Corollaries

Corollary 3.1 (Existence of local stems)
No global trunk does not mean no local structure. For each query, take the top-k tokens with highest

probability; these tokens are semantically related and form a local cluster. Their covariance matrix has a low-rank structure, so the local PC1 concentration is much higher than the random baseline.

Corollary 3.2 (Upper bound of PC1)
The theoretical upper bound of the global PC1 concentration is

$$\mathrm{PC1}_{\max} = \frac{(1+\sqrt{\gamma})^2}{d_h}$$

For GPT-2 small ($n = 31, d_h = 64$), $\gamma \approx 0.48$, and the upper bound is about 0.045.

Corollary 3.3 (Semantic Width Theorem)
The semantic dimension D_{sem} is largest in the middle layers, so PC1 is lowest there; shallow layers process lexical/positional information, with narrow semantics and high PC1; deep layers perform abstract compression, with reduced semantic dimension and rising PC1. That is,

$$\mathrm{PC1} \propto \frac{1}{D_{\mathrm{sem}}}$$

This corollary will be used in Section 5.3 for MoE predictions.

2.4 Redundancy Compensation Empirical Theorem (Theorem 4)

2.4.1 Setup

Let the model have L layers and H heads per layer. The attention output of the h-th head in the l-th layer is $A_{l,h}(X)$. Define the ablation operation $do(A_{l,h} = 0)$, and its causal effect is

$$\Delta_{l,h} = \|f(X) - f(X; \theta \setminus A_{l,h})\|$$

where f is the model output.

2.4.2 Assumptions

Assumption A5 (Linear propagation, heuristic)
When the ablation magnitude is small, the output change is approximately linear:

$$\Delta_{l,h} \approx \sum_{k=l}^{L} \gamma_k \cdot \mathrm{dep}_{k,l,h}$$

where γ_k is the contribution weight of the k-th layer to the output, and $\mathrm{dep}_{k,l,h}$ is the dependence of the k-th layer on head (l, h).

Assumption A6 (Structural head/semantic head dichotomy, heuristic)
Structural heads: high PC1, low entropy, encoding positional/syntactic information; semantic heads: low PC1, high entropy, encoding semantic information.

2.4.3 Empirical Theorem (Theorem 4)

Theorem 4 (Redundancy Compensation Empirical Theorem)
In standard Transformers, the ablation effect satisfies:

$$\Delta_{\text{Layer 0}} \gg \Delta_{\text{Layer } L-1} \gg \Delta_{\text{middle}} \gg \Delta_{\text{single head}}$$

Basis: This regularity is based on architectural intuition and ablation experiments (Section 4.7), and the experimental data support the above inequality. The theoretical mechanism is heuristic, not a strict mathematical proof. Theorem 4 does not depend on the experimental results of Assumption A1; the failure of Assumption A1 only affects the strictness of Theorem 2.

2.4.4 Supplements and Corollaries

Observation 4.1 (Exit broadcast head, empirical observation)
The key head in the exit layer (Layer $L-1$) is a "global broadcast head," whose attention target is highly concentrated on the first token of the sequence. This head is responsible for broadcasting the global representation of the first token to all positions, achieving semantic synchronization.

Corollary 4.1 (Phase transition point, empirical regularity)
When cumulatively ablating middle layers, there exists a critical point k^* such that when the number of ablated layers $k > k^*$, redundancy compensation fails and performance drops sharply. Experiments show $k^* \approx 4$.

Corollary 4.2 (Differentiation between syntax and semantics, empirical regularity)
Syntactic information relies more on shallow layers, while semantic information relies more on deep layers. Therefore, ablating shallow layers has a greater impact on syntax, and ablating deep layers has a greater impact on semantics.

2.5 Falsifiable Predictions

Based on the above results, we propose the following falsifiable predictions:

1. **Prediction 1 (Semantic width)**: The narrower the input semantic distribution, the higher the PC1 concentration, and the closer the structure is to a traditional tree with a trunk.
2. **Prediction 2 (MoE experts)**: In Mixture-of-Experts models, individual experts, due to their narrow training semantics, should have significantly higher PC1 than dense models. This prediction remains to be verified.
3. **Prediction 3 (Ablation bottlenecks)**: The effect of ablating the entry layer (Layer 0) or the exit layer (the last layer) is much larger than ablating middle layers.
4. **Prediction 4 (Functional differentiation)**: High-PC1 heads correspond to syntactic/positional functions, and low-PC1 heads correspond to semantic functions.

These predictions will be tested in the experiments of Sections 3–5.

Summary
This section establishes the complete theoretical framework of Relation Body geometry. Starting from the normalization identity of inner and outer products, we proved the Relation Body Theorem (Theorem 1), and on this basis established the Redundancy-Probability Empirical Theorem (Theorem 2), the No-Trunk Theorem (Theorem 3), and the Redundancy Compensation Empirical Theorem (Theorem 4). These results provide clear assumptions and falsifiable predictions for subsequent experiments, and offer a unified geometric language for understanding how attention organizes information.

3 Experimental Setup and Metrics

This section details the models, data, metrics, and experimental procedures used in this work. All experiments are implemented based on HuggingFace Transformers (Wolf et al., 2020). The code and data will be open-sourced upon publication.

3.1 Models and Data

3.1.1 Models

We use two scales of GPT-2 models (Radford et al., 2019) for experiments. Table 2 lists the model configurations.

Table 2: Model configurations.

Model	Layers L	Heads H	Hidden dim d	Head dim d_h
GPT-2 small	12	12	768	64
GPT-2 medium	24	16	1024	64

We choose the GPT-2 series because: (1) its architecture is standard and easy to hook and ablate; (2) it is open-source and reproducible; (3) two scales allow us to verify cross-model universality.

3.1.2 Input Text

Base text: A 31-token English paragraph used for single forward-pass analysis:

> "The cat sat on the mat. It was a sunny day and the cat wanted to play outside. The dog ran to the park and saw a bird."

Controlled-domain text: Used in Section4.8 for controlled-domain experiments, covering four semantic widths.Table 3 summarizes the four domains.

Table 3: Controlled-domain text.

Domain	Semantic width	Example content
repeat	extremely narrow	the word "the" repeated 30 times
single	narrow	pure Python code (factorial, Fibonacci, binary tree)
double	medium	mixed mathematics + poetry
multi	wide	mixed news + code + math + dialogue

All texts are encoded with the GPT-2 tokenizer without adding special tokens.

3.2 Metrics

We use six categories of metrics, each corresponding to different theoretical hypotheses.

3.2.1 Scheme A: Local Difference

Definition
For each attention head, we compute the correlation between the attention probability p_{ij} and the local difference Δ_{ij}:

$$\Delta_{ij} = \|V_j - \bar{V}_i\|$$

where

$$\bar{V}_i = \sum_j p_{ij} V_j$$

is the context-weighted average of the current query i.

Metric

$$r_A = \text{corr}(p_{ij}, \Delta_{ij})$$

We compute the Pearson correlation coefficient over all causally valid (i, j) pairs.

Interpretation

- $r_A < 0$: high-probability tokens converge to the context mean, i.e., local redundancy;
- $r_A \to -1$: strong local redundancy;
- $r_A \to 0$: no local structure.

Corresponding theoretical result: Theorem 2 (Redundancy-Probability Empirical Theorem).

3.2.2 Scheme B: Global Principal Component Energy

Definition
For the entire sequence's Value matrix $V \in \mathbb{R}^{n \times d_h}$, we perform SVD:

$$V = USV^T$$

We take the top $K = 5$ principal components $V^T[:5]$. For each token j, we compute its energy proportion in the top 5 principal component directions:

$$\text{energy}_j = \frac{\|\text{proj}_{top5}(V_j)\|^2}{\|V_j\|^2}$$

Metric

$$r_B = \text{corr}(\text{avg_}p_j, \text{energy}_j)$$

where $\text{avg_}p_j$ is the average probability of token j being attended to as a key by all queries.

Interpretation

- $r_B > 0$: high-probability tokens are concentrated in global principal components, indicating a global trunk;
- $r_B < 0$: high-probability tokens lie outside global principal components, indicating no trunk.

Corresponding theoretical result: Theorem 3 (No-Trunk Theorem).

3.2.3 Scheme C: Global Residual Norm

Definition
Same SVD decomposition as in Scheme B. For each token j, compute its residual norm:

$$\text{residual}_j = \|V_j - \text{proj}_{top5}(V_j)\|$$

Metric

$$r_C = \text{corr}(\text{avg_}p_j, \text{residual}_j)$$

Interpretation

- $r_C < 0$: high-probability tokens have small residuals, fully explained by global structure;
- $r_C > 0$: high-probability tokens have large residuals, not explained by global structure.

Corresponding theoretical result: Theorem 3 (No-Trunk Theorem).

3.2.4 Local PCA: Local Stem Strength

Definition
For each query i, take the top-k tokens with highest attention probability (we use $k = 10$), and extract their Value vectors $V_{\text{top}} \in \mathbb{R}^{k \times d_h}$. After centering, perform SVD.Define the local principal component analysis (PCA) concentration:

$$V_{\text{top}} - \bar{V}_{\text{top}} = USV^T$$

Define the local PC1 concentration:

$$\text{conc}_i = \frac{S_1^2}{\sum_k S_k^2}$$

Metric
Average over all queries:

$$\text{mean_conc} = \frac{1}{n} \sum_{i=1}^{n} \text{conc}_i$$

Interpretation

- conc $\to 1$: the top-k Values are almost collinear, strong local stem;
- conc $\to 1/d_h$: completely random and dispersed, no local structure;
- Experiments observe a mean around 0.34, far above the random baseline 0.0156.

Corresponding theoretical result: Corollary 3.1 (Existence of local stems).

3.2.5 Ablation Effect

Definition
For a specified head or layer, we ablate it (set its output to zero) and measure the performance change on the test set:

$$\Delta = \text{score}_{\text{after}} - \text{score}_{\text{baseline}}$$

Test sets

- **Syntactic test set**: 60 questions covering subject-verb agreement, tense, pronouns, articles, and prepositions;
- **Semantic test set**: 60 questions covering word sense selection and commonsense reasoning.

Scoring function
We use the log-probability difference as a continuous metric:

$$\text{score} = \log P(\text{correct} \mid \text{context}) - \log P(\text{wrong} \mid \text{context})$$

Ablation granularity

1. Single-head ablation: zero out the output of a single attention head;
2. Layer ablation: zero out the outputs of all heads in a layer;
3. Cumulative ablation: progressively ablate multiple middle layers from small to large indices;
4. Activation patching: replace the output of a specified head with its mean across all tokens.

Corresponding theoretical result: Theorem 4 (Redundancy Compensation Empirical Theorem).

3.2.6 Controlled-Domain Experiment Metrics

Definition
For each input domain, repeat all the above metrics (Schemes A/B/C, local PCA) to obtain the average PC1 concentration for that domain.

Core test
Verify monotonicity:

$$\text{PC1(repeat)} > \text{PC1(single)} > \text{PC1(double)} > \text{PC1(multi)}$$

Corresponding theoretical result: Corollary 3.3 (Semantic Width Theorem).

3.2.7 Cross-Model Validation

To assess whether the observed geometric patterns are model-specific or universal, we repeat the metrics defined above (Schemes A/B/C, local PCA, and attention entropy) on both GPT-2 small and GPT-2 medium. We compare layer-wise PC1 concentration, the cross-model r_A distribution, and head-level statistics. The results of this validation are reported in Section 4.4.

3.2.8 Functional Analysis of High-PC1 Heads

To characterize the functional roles of heads with extreme PC1 values, we select the head with the highest PC1 (Layer 1, Head 11 in GPT-2 small) and a representative low-PC1 head (Layer 3, Head 1). For each selected head, we compute the mean attention entropy, mean top-1 probability, and the top-1 target distribution across queries. We also inspect the attention matrix qualitatively to identify positional or syntactic patterns. This analysis is reported in Section 4.5.

3.2.9 Layer 11 Semantic Exit Analysis

To investigate the role of the final layer in semantic aggregation, we perform per-head ablation on Layer 11 and measure the semantic Δ for each head. We identify the head with the largest semantic contribution and analyze its attention target distribution. We further compare Layer 0 (entry) and Layer 11 (exit) in terms of mean PC1 and mean attention entropy. This analysis is reported in Section 4.6.

3.3 Experimental Procedure

All experiments follow a unified procedure:

1. **Load model**: Use `AutoModelForCausalLM` and `AutoTokenizer`;
2. **Register hooks**: Register forward hooks on each layer's `attn` module to extract Q/K/V;
3. **Forward pass**: Input text and collect Q/K/V from all layers;
4. **Compute metrics**: Compute Schemes A/B/C, local PCA, entropy, etc., for each layer and head;
5. **Ablation experiments**: Register ablation hooks, repeat forward pass, and compute performance changes;
6. **Cumulative ablation**: Gradually increase the number of ablated layers and record the performance curve;
7. **Controlled-domain experiments**: Change input text and repeat steps 3–4;
8. **Visualization**: Generate heatmaps, scatter plots, and layer trend plots.

Randomness control:

- All experiments use `model.eval()` mode;
- No training is involved, so there are no random seed issues;
- For ablation experiments, the same input text and test sets are used.

Computational resources:

- All experiments run on a single GPU (e.g., NVIDIA Theorem 4 or V100);
- A single experiment takes seconds to minutes, and full reproduction takes about 1 hour.

3.4 Summary

This section defines six categories of metrics, each corresponding to different theoretical predictions in Relation Body geometry. Schemes A/B/C are used to test local redundancy and global trunk, local PCA quantifies local stem strength, ablation experiments verify the redundancy compensation theorem, and controlled-domain experiments directly manipulate the semantic width variable. All metrics are based on a single forward pass or simple ablation, making them computationally efficient and reproducible.

The next section presents the experimental results of these metrics on GPT-2 small and GPT-2 medium.

4 Experimental Results

This section presents all experimental results based on the metrics defined in Section 3. The experiments are organized in logical order: first, we verify the universality of local redundancy (Section 4.1); second, we test the global trunk hypothesis (Section 4.2); third, we quantify local stem strength (Section 4.3); fourth, we perform cross-model validation (Section 4.4). On this basis, we further analyze the functions of high-PC1 heads (Section 4.5), the semantic exit mechanism in Layer 11 (Section 4.6), ablation and redundancy compensation (Section 4.7), and finally provide the core causal evidence from controlled-domain experiments (Section 4.8).

4.1 Universality of Local Redundancy (Scheme A)

4.1.1 GPT-2 small

For all 144 (layer, head) pairs of GPT-2 small, we compute the correlation coefficient r_A of Scheme A.Table 4 summarizes the statistics.

Table 4: GPT-2 small: Scheme A.

Metric	Value
Mean r_A	**-0.6783**
Negative correlation ratio	**100.00%**
Maximum negative correlation	-0.9333 (Layer 8, Head 1)
Minimum negative correlation	-0.2242 (Layer 11, Head 8)

Conclusion: All layers and all heads exhibit local redundancy, i.e., high attention probability corresponds to low local difference. This verifies hypothesis Theorem 2.

4.1.2 GPT-2 medium

For all 384 (layer, head) pairs of GPT-2 medium, we compute Scheme A.Table 5 summarizes the statistics.

Table 5: GPT-2 medium: Scheme A.

Metric	Value
Mean r_A	**-0.6657**
Negative correlation ratio	**99.74%**
Maximum negative correlation	-0.9482 (Layer 9, Head 9)
Minimum negative correlation	+0.0780 (Layer 1, Head 15)

Conclusion: GPT-2 medium is almost identical to small, indicating that local redundancy is a universal property of Transformers and does not change with model scale.

4.1.3 Summary

High probability ⇒ high local redundancy ($r_A < 0$) is a universal regularity across models, layers, and heads.

Figure 2a shows the scatter plot of Scheme A (local difference vs. attention probability). The cross-model r_A distribution is shown in Figure A2 (Appendix C).

4.2 Testing the Global Trunk Hypothesis (Schemes B/C)

4.2.1 Scheme B: Global Principal Component Energy

Table 6 reports the correlation between high-probability tokens and global principal component energy.

Table 6: Scheme B: global principal component energy.

Model	Mean r_B	Positive correlation ratio
GPT-2 small	-0.3129	16.67%
GPT-2 medium	-0.3578	12.50%

Conclusion: The energy proportion of high-probability tokens in global principal components decreases rather than increases. Hypothesis Theorem 3 (existence of a global trunk) is rejected.

Figure 2b shows the scatter plot of Scheme B (global principal component energy vs. average attention probability).

4.2.2 Scheme C: Global Residual Norm

Table 7 reports the correlation between high-probability tokens and global residual norm.

Table 7: Scheme C: global residual norm.

Model	Mean r_C	Negative correlation ratio
GPT-2 small	-0.4006	89.58%
GPT-2 medium	-0.4213	91.15%

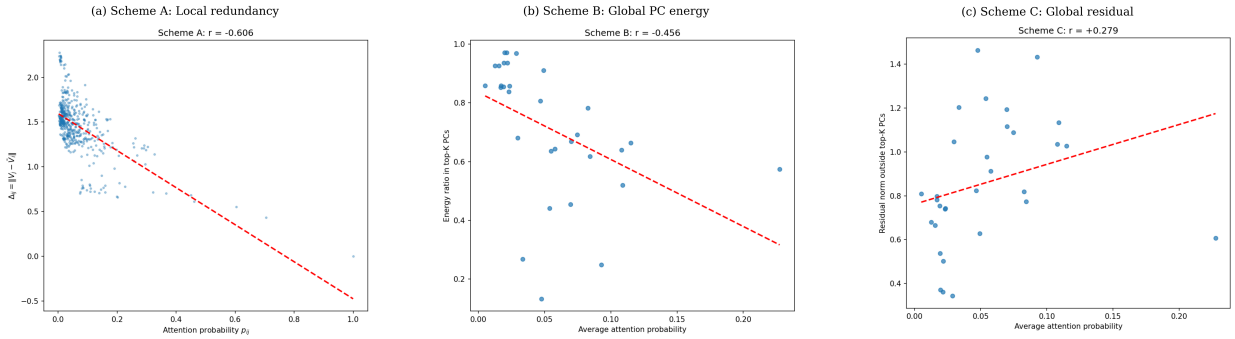
Conclusion: High-probability tokens have relatively small residuals and are partially explained by global principal components. However, since Scheme B already rejects the global trunk, the negative correlation of Scheme C only indicates that global principal components can partially cover high-probability tokens, but are insufficient to constitute a "trunk".

Figure 2c shows the scatter plot of Scheme C (global residual norm vs. average attention probability).

4.2.3 Summary

No global trunk: energy is dispersed, with no dominant direction spanning the entire sequence.

Global principal components can only partially explain high-probability tokens, and are insufficient to form a "trunk".



(a) Scheme A: local difference vs. attention probability.

(b) Scheme B: global principal component energy vs. average attention probability.

(c) Scheme C: global residual norm vs. average attention probability.

Figure 2: Three global metrics for attention heads.

4.3 Local Stem Strength (Local PCA)

4.3.1 GPT-2 small

For each query, take the top-10 tokens with highest probability and compute the PC1 concentration.Table 8 summarizes the statistics for GPT-2 small.

Table 8: GPT-2 small: local PCA.

Metric	Value
Mean PC1	**0.3436**
Random baseline $1/d_h$	0.0156
Maximum	0.6829 (Layer 1, Head 11)
Minimum	0.2588
PC1 > 0.3 ratio	82.64%
PC1 > 0.5 ratio	2.08%
Mean $r(\text{conc}, \text{entropy})$	**-0.4058**

4.3.2 GPT-2 medium

For GPT-2 medium, we repeat the same analysis. Table 9 summarizes the statistics.

Table 9: GPT-2 medium: local PCA.

Metric	Value
Mean PC1	**0.3595**
Random baseline $1/d_h$	0.0156
Maximum	0.7523 (Layer 23, Head 4)
Minimum	0.2588
PC1 > 0.3 ratio	91.15%
PC1 > 0.5 ratio	8.33%
Mean $r(\text{conc}, \text{entropy})$	**-0.4271**

4.3.3 Summary

Local stems exist: mean PC1 is 0.34–0.36, far above the random baseline.

But they are not thick trunks: PC1 does not exceed 0.8, and most heads lie between 0.3 and 0.5.

The more concentrated the attention (lower entropy), the stronger the local stem (higher PC1).

The complete PC1 layer $\times$ head distribution is shown in **Appendix FigureA1** (GPT-2 medium).

4.4 Cross-Model Validation

4.4.1 Consistency of Layer Trends

Table 10 compares the layer-wise PC1 concentration between GPT-2 small and GPT-2 medium.

Table 10: Consistency of layer trends.

Model	Shallow PC1 (0–2)	Middle PC1 (3–10)	Deep PC1 (11+)
GPT-2 small	0.40	0.32	0.35
GPT-2 medium	0.47	0.32	0.43

Conclusion: Both models exhibit a U-shaped curve: high in shallow layers, low in middle layers, and rising again in deep layers.

4.4.2 PC1 Is Independent of Model Scale

Table 11 shows that PC1 concentration is almost independent of model scale.

Table 11: PC1 is independent of model scale.

Model	head_dim	Mean PC1
GPT-2 small	64	0.3436
GPT-2 medium	64	0.3595

Conclusion: PC1 concentration is determined by head_dim and is almost independent of the number of layers, heads, or parameters.

4.4.3 Summary

Local stem strength is determined by head_dim and is independent of model scale.

The layer trend (strong shallow, weak middle, rising deep) is a universal regularity of Transformers.

The cross-model layer trend comparison is shown in **Appendix FigureA5**; the cross-model r_A distribution comparison is shown in **Appendix FigureA2**.

4.5 Functional Analysis of High-PC1 Heads

4.5.1 High-PC1 Head: Layer 1, Head 11

Table 12 summarizes the metrics for the high-PC1 head (Layer 1, Head 11).

Table 12: High-PC1 head: Layer 1, Head 11.

Metric	Value
PC1	**0.6829**
Mean entropy	**1.0192**
Mean top-1 probability	**0.6954**
Top-1 target distribution	self + first token "The" + punctuation

Figure 3 shows the attention matrix of this head. Its behavioral characteristics are:

- The attention matrix shows a strong diagonal (attending to itself);
- At periods, it jumps to the first token "The";
- A few cross-word attentions (e.g., "cat" attends to "The").

Functional label: **positional anchoring head + sentence-boundary head**.

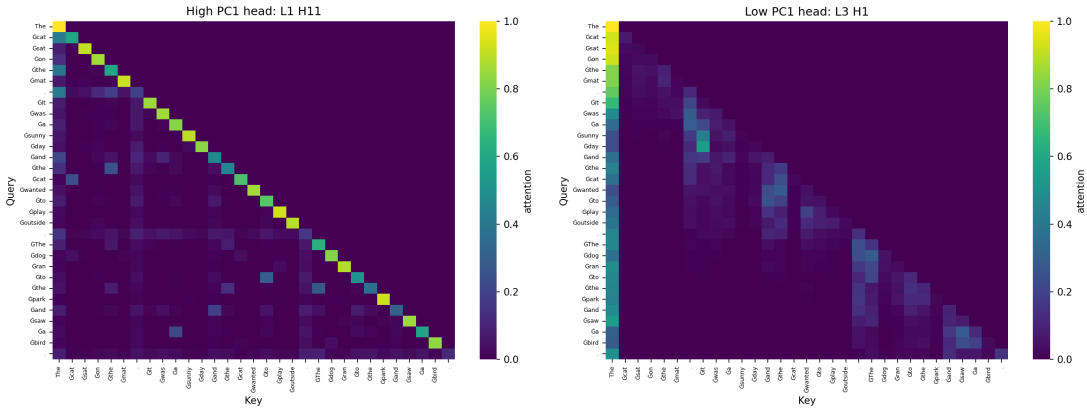


Figure 3: Attention matrix comparison between a high-PC1 head (L1 H11) and a low-PC1 head (L3 H1).

4.5.2 Low-PC1 Head: Layer 3, Head 1

Table 13 summarizes the metrics for the low-PC1 head (Layer 3, Head 1).

Table 13: Low-PC1 head: Layer 3, Head 1.

Metric	Value
PC1	**0.2899**
Mean entropy	**1.4071**
Behavioral characteristics	diffuse attention, no fixed pattern

Functional label: **semantic head (general, diffuse)**.

4.5.3 Comparison

Table 14 compares the two heads.

Table 14: Comparison of high- and low-PC1 heads.

Head	PC1	Entropy	Function
L1 H11	0.68	1.02	positional/syntactic
L3 H1	0.29	1.41	semantic

Entropy difference: $1.4071 - 1.0192 = 0.3879$.

4.5.4 Summary

High-PC1 heads = low entropy = structural heads (syntactic/positional).

Low-PC1 heads = high entropy = semantic heads (general/diffuse).

4.6 Layer 11 Semantic Exit and Apical Dominance

4.6.1 Per-Head Ablation

We ablate each of the 12 heads in Layer 11 and measure the semantic Δ.Table 15 shows the per-head ablation results in Layer 11.

Table 15: Per-head ablation in Layer 11.

Head	Semantic Δ	PC1	Entropy	Rank
7	**-0.1422**	0.324	1.030	1
5	-0.0683	0.292	0.790	2
1	-0.0587	0.296	1.301	3
4	-0.0563	0.340	1.261	4
8	-0.0048	**0.546**	2.010	11

4.6.2 Attention Target of Head 7

Top-1 target distribution (Head 7): 'The': 31 times (100.0%)

All 31 tokens have the first token "The" as their top-1 target.

Figure 4 shows the attention matrix of Head 7, which 100% attends to the first token.

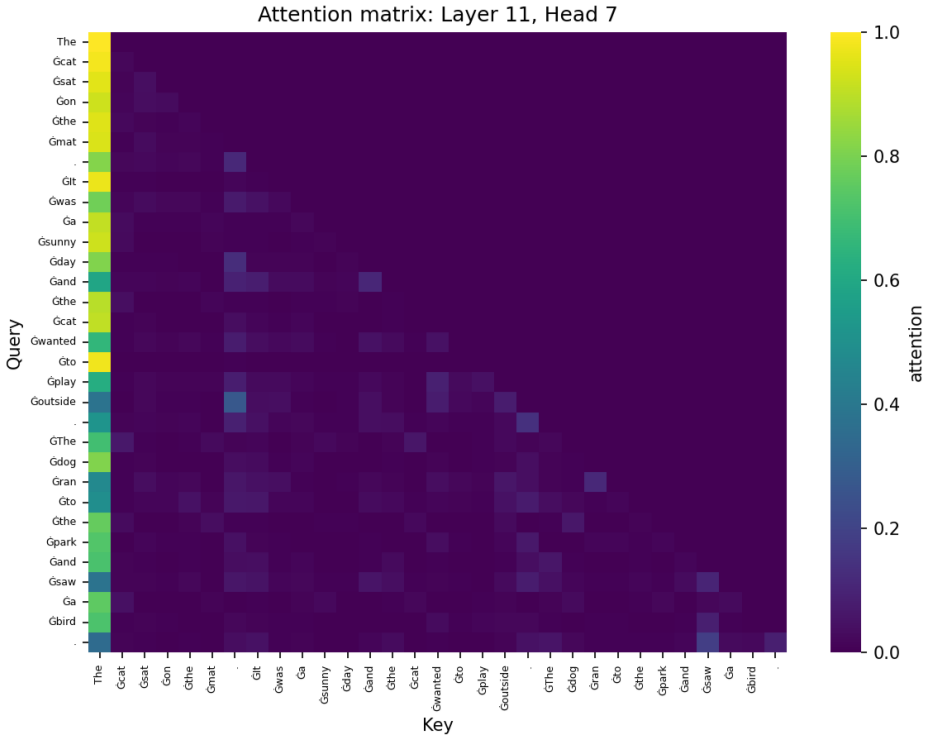


Figure 4: Attention matrix heatmap of Layer 11 Head 7, which 100% attends to the first token.

4.6.3 Functional Definition

Head 7 = global semantic broadcast head: extracts a global representation from the first token and broadcasts it to all positions.

Analogy: apical dominance in plants—the first token is the apical bud, Head 7 is auxin, and all positions converge toward the first token.

4.6.4 Comparison between Layer 0 and Layer 11

Table 16 compares Layer 0 (entry) and Layer 11 (exit).The corresponding scatter plot is provided in Appendix C (Figure A7).

Table 16: Comparison between Layer 0 and Layer 11.

Metric	Layer 0 (entry)	Layer 11 (exit)
Mean PC1	0.380	0.351
Mean entropy	1.739	1.338
Highest PC1 head	H10 (0.484)	H8 (0.546)
Semantically critical head	—	H7 (PC1=0.324)

Conclusion: The entry is divergent, while the exit is focused; the critical head at the exit has moderate PC1 but is functionally irreplaceable.

4.6.5 Summary

Layer 11 Head 7 = global semantic broadcast head, 100% attending to the first token.

High PC1 = structural redundancy; moderate PC1 = semantic criticality.

4.7 Ablation and Redundancy Compensation

4.7.1 Single-Head Ablation

For representative heads, single-head ablation effects are all within ± 0.025. Table 17 summarizes the results.

Table 17: Single-head ablation.

Head	Syntactic Δ	Semantic Δ
L1 H11	$+0.0116$	-0.0010
L3 H1	-0.0120	-0.0121
L5 H5	$+0.0048$	-0.0078
L11 H8	$+0.0249$	-0.0197

Conclusion: Single-head ablation is fully compensated by multi-head redundancy.

4.7.2 Layer Ablation

Table 18 shows the layer ablation results.

Table 18: Layer ablation.

Layer	Syntactic Δ	Semantic Δ	Interpretation
L0	**-2.6228**	**-3.5400**	information entry bottleneck
L1	-0.1152	-0.0359	minor
L2–L10	within ±0.2	within ±0.2	redundancy compensation
L11	-0.1091	**-0.7619**	semantic exit

Conclusion: Entry and exit are bottlenecks, while middle layers are highly redundant.

4.7.3 Cumulative Ablation

Table 19 shows the cumulative ablation results.

Table 19: Cumulative ablation.

k	Ablated layers	Syntactic Δ	Semantic Δ
0	none	0	0
3	[1,2,3]	-0.32	-0.23
4	**[1,2,3,4]**	**-0.97**	**-0.83**
5	[1–5]	-1.09	-1.24
10	[1–10]	-1.85	-3.63

Figure 5 shows the cumulative ablation curves and the phase transition point $k^* = 4$.

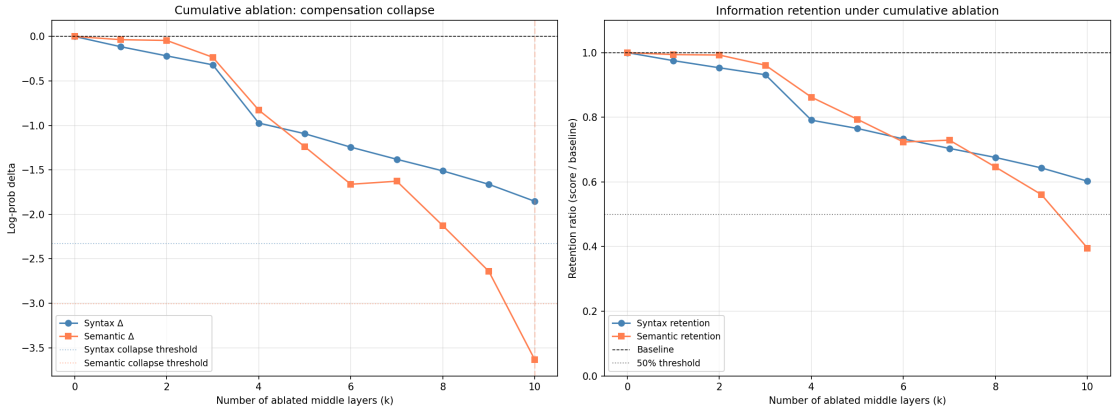


Figure 5: Cumulative ablation curves and the phase transition point $k^* = 4$.

Phase transition point: $k^* = 4$, after which redundancy compensation fails and performance drops sharply.

4.7.4 Summary

$$\Delta_{\text{Layer 0}} \gg \Delta_{\text{Layer 11}} \gg \Delta_{\text{middle}} \gg \Delta_{\text{single head}}$$

Cumulative ablation shows a phase transition at $k^* = 4$: redundancy compensation has an upper limit.

Table 20 summarizes the ablation effects across all granularities.

Table 20: Summary of ablation effects.

Ablation granularity	Syntactic Δ	Semantic Δ	Remark
Single-head ablation	±0.025	±0.025	compensated by multi-head redundancy
Middle-layer ablation (L2–L10)	±0.2	±0.2	compensated by other layers
Entry-layer ablation (L0)	−2.62	−3.54	information entry bottleneck
Exit-layer ablation (L11)	−0.11	−0.76	semantic exit bottleneck
Cumulative ablation $k = 4$	−0.97	−0.83	phase transition point
Cumulative ablation $k = 10$	−1.85	−3.63	redundancy exhausted

4.8 Controlled-Domain Experiments

4.8.1 PC1 vs. Semantic Width

Table 21 summarizes the mean PC1 concentration for the four semantic widths.

Table 21: PC1 vs. semantic width.

Domain	Semantic width	Mean PC1	Std
repeat	extremely narrow	**0.7637**	0.1147
single	narrow	**0.3773**	0.0637
double	medium	**0.3231**	0.0569
multi	wide	**0.3079**	0.0533

4.8.2 Monotonicity Test

Monotonically decreasing: True Hypothesis SUPPORTED

$$\boxed{\text{PC1(repeat)} > \text{PC1(single)} > \text{PC1(double)} > \text{PC1(multi)}}$$

4.8.3 Layer Trends

- **repeat**: PC1 in all layers ranges from 0.6 to 0.9, highest in shallow layers;
- **single/double/multi**: PC1 ranges from 0.3 to 0.45, exhibiting a U-shape;
- The three wide-domain curves almost overlap.

4.8.4 r_A Distribution

- The mean r_A across all domains is approximately -0.46;
- The repeat distribution is the widest (from -1.0 to +0.4);
- The other three domains are concentrated between -0.6 and -0.2.

Figure 6 summarizes all four analyses above.

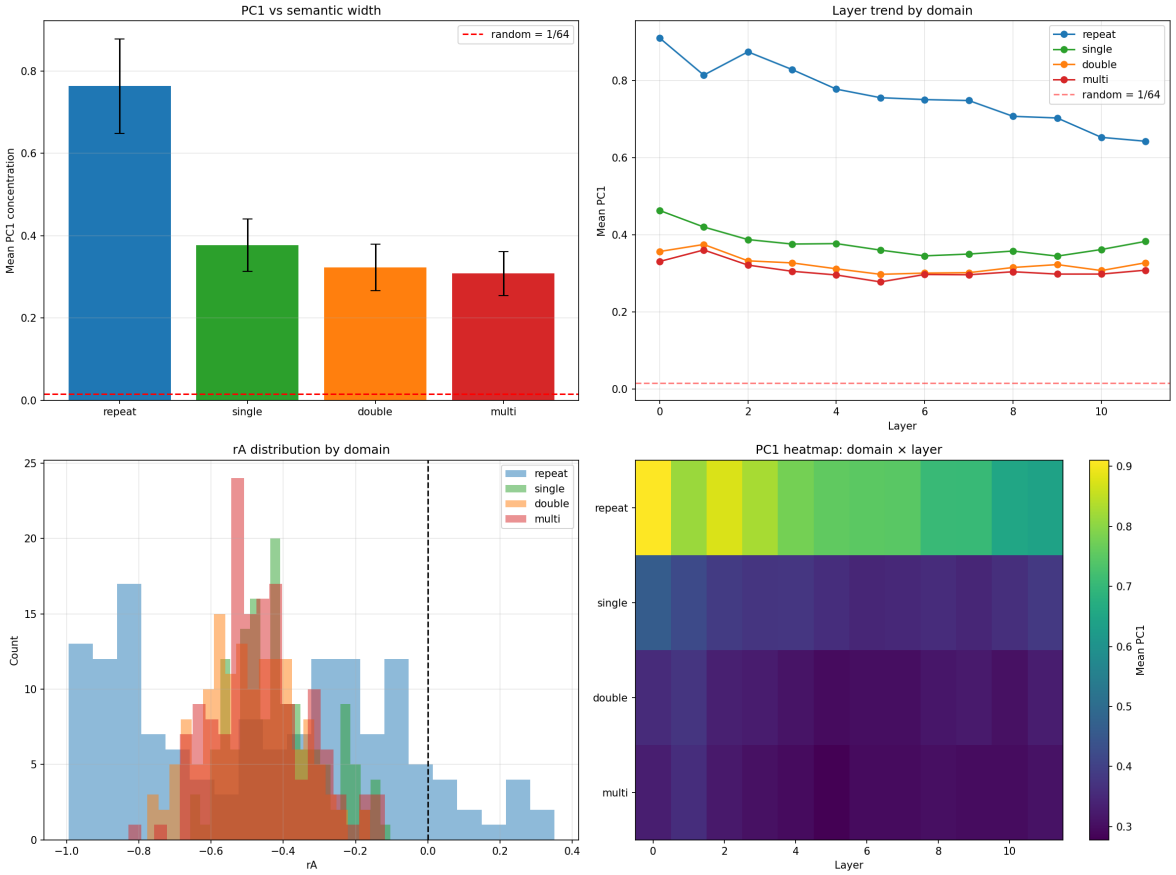


Figure 6: PC1 vs. semantic width bar chart, layer trends, r_A distribution, and PC1 heatmap.

Conclusion: Local redundancy is an architectural property and is independent of semantic width.

4.8.5 Summary

Narrower semantics ⇒ higher PC1 ⇒ closer to a traditional tree with a trunk.

Dense models (wide semantics) ⇒ PC1 ≈ 0.31˘0.34 ⇒ shrub.

4.9 Overview of Experimental Results

Table 22 summarizes the correspondence between each experiment and the theoretical results.

Table 22: Overview of experimental results.

Experiment	Core finding	Corresponding result	Status
4.1 Local redundancy	100% negative correlation	Theorem 2	Yes
4.2 No global trunk	r_B negative correlation	Theorem 3	Yes
4.3 Local PCA	PC1 = 0.34, far above random	Corollary 3.1	Yes
4.4 Cross-model	PC1 determined by head_dim	Theorem 3	Yes
4.5 High-PC1 heads	structural heads (low entropy)	Corollary 2.1	Yes
4.6 Layer 11	global semantic broadcast head	Observation 4.1	Yes
4.7 Ablation	entry/exit bottlenecks, phase transition $k^* = 4$	Theorem 4	Yes
4.8 Controlled domain	narrower semantics → higher PC1	Corollary 3.3	Yes

Core conclusion:

Attention mechanism = no global trunk + local stems + inter-head functional differentiation + inter-layer redundancy compensation.

Semantic width is the causal determinant of PC1.

Dense models are shrubs; MoE experts are predicted to be traditional trees.

5 Discussion

This section integrates the experimental findings from Section 4, proposes the "no-trunk shrub" model, discusses its relationship with existing interpretability research, provides falsifiable predictions for Mixture-of-Experts (MoE) models, and outlines the limitations and future directions of this work.

5.1 The No-Trunk Shrub Model

Combining all experimental results from Section 4, we can summarize the geometric structure of attention in trained Transformers as follows:

no global trunk + local stems + inter-head functional differentiation + inter-layer redundancy compensation

This structure is called the "no-trunk shrub."

5.1.1 No Global Trunk

Schemes B and C show that the energy proportion of high-probability tokens in global principal components decreases rather than increases, and the global singular value spectrum is close to the Marchenko-Pastur distribution with no isolated large eigenvalues. This means there is no "thick trunk" spanning the entire sequence.

Wide semantic distribution $\Rightarrow$ PC1 $\to 1/d_h \Rightarrow$ no trunk.

5.1.2 Local Stems

Local PCA (Section 4.3) shows that for each query, the Value vectors of the top-k high-probability tokens have a dominant direction (mean PC1 around 0.34, far above the random baseline 0.0156). This corresponds to "local stems"—local structural centers.

$$\boxed{\text{Local semantic clustering} \Rightarrow \text{PC1} \gg 1/d_h \Rightarrow \text{local stems.}}$$

5.1.3 Inter-Head Functional Differentiation

Sections 4.5 and 4.6 show that:

- **High-PC1 heads**: low entropy, specialized, corresponding to syntactic/positional functions (e.g., L1 H11);
- **Low-PC1 heads**: high entropy, general, corresponding to semantic functions (e.g., L3 H1);
- **Layer 11 Head 7**: moderate PC1 (0.324), but with the largest semantic contribution, functioning as a "global semantic broadcast head."

$$\boxed{\text{High PC1} = \text{structural redundancy; low PC1} = \text{semantic diversity; moderate PC1} = \text{critical broadcast.}}$$

5.1.4 Inter-Layer Redundancy Compensation

Ablation experiments in Section 4.7 show that:

- **Single-head ablation**: effect ± 0.025, almost no impact;
- **Middle-layer ablation**: effect ± 0.2, compensated by other layers;
- **Entry-layer (L0) ablation**: effect -2.6 (syntax), -3.5 (semantics), information bottleneck;
- **Exit-layer (L11) ablation**: semantic $\Delta = -0.76$, semantic exit bottleneck;
- **Cumulative ablation**: phase transition at $k^* = 4$, redundancy compensation has an upper limit.

$$\boxed{\text{The entry/exit are bottlenecks, middle layers are redundant shrubs, and the phase transition point is } k^* = 4.}$$

5.2 Relationship with Existing Interpretability Research

5.2.1 Functional Classification of Attention Heads

Existing research has identified various functional heads: positional heads, syntactic heads (Tenney et al., 2019), induction heads (Olsson et al., 2022), and anaphora heads (Clark et al., 2019; Voita et al., 2019). This work provides a unified geometric explanation for these functional labels:

$$\boxed{\text{Functional head} = \text{a specific region on the Relation Body (high } c \text{ or high } s\text{).}}$$

- High-PC1 heads: corresponding to high c (same/parallel), low entropy, structural function;
- Low-PC1 heads: corresponding to high s (different/orthogonal), high entropy, semantic function.

5.2.2 Sparse Autoencoders (SAE)

Sparse autoencoders (Bricken et al., 2023) decomposes neuron activations through dictionary learning to find interpretable features. The Relation Body geometry in this paper can be viewed as the geometric dual of SAE:

SAE seeks sparse directions; Relation Body seeks redundancy-independence balance.

Both attempt to reveal the structure of hidden representations, but this paper starts from the first principles of inner/outer products and provides a more fundamental geometric language.

5.2.3 Causal Tracing and Ablation

The ablation experiments in Section 4.7 are consistent with causal tracing methods (Meng et al., 2022) , but further quantify:

- Entry/exit bottlenecks;
- Middle-layer redundancy compensation;
- Phase transition point $k^* = 4$.

These results provide a new theoretical perspective for causal tracing.

5.2.4 Relationship with "Induction Heads"

Induction heads (Olsson et al., 2022) are responsible for pattern completion and are central to in-context learning(ICL). In the Relation Body framework, induction heads correspond to:

High c (same) + high s (different) $\Rightarrow$ strong pattern matching.

Their PC1 may lie between structural heads and semantic heads, forming a "diagonal region" on the Relation Body.

5.3 Falsifiable Predictions for MoE

Based on Corollary 3.3 (Semantic Width Theorem), we propose the following core prediction for Mixture-of-Experts (MoE) models (Shazeer et al., 2017; Fedus et al., 2022):

The PC1 of individual experts should be significantly higher than that of dense models.

This prediction remains to be empirically verified.

5.3.1 Theoretical Basis

In MoE, each expert processes only a specific subdomain (e.g., mathematics, code, French), and its training semantic distribution D_{sem} is much narrower than that of dense models. By Corollary 3.3:

$$\text{PC1} \propto \frac{1}{D_{\text{sem}}}$$

Therefore:

Narrow expert semantics $\Rightarrow$ high PC1 $\Rightarrow$ closer to a traditional tree with a trunk.

5.3.2 Predicted Values

Table 23 shows the predicted values for MoE experts.

Table 23: Predicted values for MoE experts.

Model type	Expected PC1	Expected structure
Dense model (GPT-2)	0.31–0.36	shrub
Individual expert	> 0.5	traditional tree (with trunk)

5.3.3 Verification Plan

1. Load an open-source MoE model (e.g., Mixtral, Qwen-MoE);
2. Hook the routing layer and mark the expert assignment of each token;
3. Collect QKV separately for each expert;
4. Repeat the global PCA + local PCA analysis of this paper for each expert;
5. Compare the PC1 of dense models and experts.

If the expert PC1 is significantly higher than that of dense models, Corollary 3.3 (Semantic Width Theorem) is supported; if the expert PC1 remains around 0.34, the Prediction 2 is rejected.

5.3.4 Further Predictions

- The PC1 of experts should be negatively correlated with their training data width;
- The inter-head variance within experts should be smaller than that of dense models;
- The layer trend of experts should be flatter (because semantics are narrow, PC1 does not change drastically across layers).

5.4 "Passive Emergence" under Causal Masking (Explanatory Observation)

Note: This section is an explanatory observation of experimental phenomena, not a strict theorem. Its argument is based on architectural intuition and experimental evidence, and does not constitute a mathematical proof.

Our experiments reveal an important architectural asymmetry:

Causal masking prohibits backward broadcast, causing global semantics to only "passively emerge."

In standard GPT-2:

- **First 11 layers**: each token independently evolves, with no global synchronization;
- **First token**: passively accumulates information through the residual stream, unable to actively integrate;
- **Layer 12**: L11 H7 lets all tokens read the first token, achieving "pseudo-global synchronization."

This finding explains why:

- L11 H7 must exist (to compensate for the missing backward broadcast);
- L11 H7 has the largest ablation effect (semantic $\Delta = -0.14$);
- The global representation of causal Transformers is "emergent" rather than "constructed."

Standard Transformer: pseudo-global (passive accumulation + terminal broadcast).

This observation also motivates future work, such as bidirectional broadcast models that break causal masking.

5.5 Limitations

This work has the following limitations:

1. **Model scope**: Experiments are mainly conducted on GPT-2 small and medium, and have not yet been validated on larger models (e.g., LLaMA (Touvron et al., 2023), Qwen) or MoE (Shazeer et al., 2017; Fedus et al., 2022)

2. **Sequence length**: Input text length is limited (31–150 tokens); longer sequences may exhibit different regularities.

3. **Controlled-domain experiments**: repeat/single/double/multi are artificially constructed, not natural training distributions.

4. **Causal masking**: This paper only analyzes standard causal Transformers and does not involve bidirectional attention (e.g., BERT (Devlin et al., 2019)).

5. **Theoretical assumptions**: The QKV alignment assumption of Theorem 2 and the random projection assumption of Theorem 3 are approximations and require more empirical validation.

6. **MoE prediction**: Has not yet been verified on real MoE models.

Series context: A notable limitation of the current work is its empirical focus on the GPT-2 model family. To address this, the present paper serves as the foundational theoretical work of a two-paper series. A companion paper will extend the Relation Body framework to cross-model generalization (e.g., LLaMA, Qwen) and alignment paradigms, for which the theorems established here (Theorems 1–4) provide the necessary mathematical foundation.

5.6 Future Work

Based on the findings of this paper, future work can proceed in the following directions:

1. **MoE validation**: Test whether expert PC1 is significantly higher than that of dense models on real MoE models.

2. **Larger models**: Repeat the experiments in this paper on LLaMA, Qwen, etc., to verify the universality of the regularities.

3. **Bidirectional broadcast models**: Design architectures that break causal masking and allow backward broadcast, and test whether they can actively construct global representations.

4. **Self-organized growth models**: Based on Relation Body geometry, design generation paths that do not preset QKV/softmax, and observe the spontaneous emergence of information trees (see a companion paper).

5. **Geometric interpretability toolkit**: Package the metrics of this paper (Schemes A/B/C, local PCA, ablation) into open-source tools for the community.

6. **Theoretical deepening**: Strictly prove the Marchenko-Pastur approximation of Theorem 3, relax Assumption A4, analytically derive the phase transition point k^* of Theorem 4, and extend Relation Body geometry to a unified framework for multi-head, multi-layer, and MoE models.

5.7 Summary

Attention mechanism = no global trunk + local stems + inter-head functional differentiation + inter-layer redundancy compensation.

Semantic width is the causal determinant of PC1.

Dense models are shrubs; MoE experts are predicted to be traditional trees.

Causal masking causes global semantics to "passively emerge" rather than be "actively constructed."

These conclusions provide a unified language for understanding the internal information geometry of Transformers and offer testable directions for the design of next-generation architectures.

6 Conclusion and Future Work

6.1 Conclusion

Starting from the fundamental identity of inner and outer products, this paper establishes a unified geometric framework—Relation Body geometry—for understanding how Transformer attention mechanisms organize information. The main conclusions are as follows.

6.1.1 Theoretical Contributions

We proved the Relation Body Theorem (Theorem 1): the relationship between any two non-zero vectors can be normalized to a point on the unit circle, satisfying $c^2 + s^2 = 1$. On this basis, we established three core results:

- **Theorem 2 (Redundancy-Probability Empirical Theorem)**: high attention probability corresponds to high local redundancy;
- **Theorem 3 (No-Trunk Theorem)**: wide semantic distributions cause the global PC1 to approach the random baseline, and no trunk spans the entire sequence;
- **Theorem 4 (Redundancy Compensation Empirical Theorem)**: entry and exit layers are information bottlenecks, while middle layers possess strong redundancy compensation.

These results jointly characterize the geometric nature of attention mechanisms:

Attention mechanism = information organization process on the Relation Body.

6.1.2 Experimental Findings

Systematic experiments on GPT-2 small and GPT-2 medium reveal the following regularities:

1. **Universality of local redundancy**: all layers and all heads exhibit $r_A < 0$ (100% negative correlation), with high-probability tokens converging to the context mean;
2. **No global trunk**: Schemes B/C show that global principal components cannot explain high-probability tokens, and energy is dispersed;
3. **Local stems**: local PCA mean is about 0.34, far above the random baseline 0.0156;
4. **Inter-head functional differentiation**: high-PC1 heads correspond to syntactic/positional functions (low entropy), while low-PC1 heads correspond to semantic functions (high entropy);

5. **Semantic exit broadcast**: Layer 11 Head 7 is a global semantic broadcast head, 100% attending to the first token, and its function is irreplaceable;
6. **Redundancy compensation and phase transition**: cumulative ablation exhibits a phase transition at $k^* = 4$, and redundancy compensation has an upper limit;
7. **Controlled-domain causal verification**: the narrower the input semantics, the higher the PC1, and the monotonic decreasing relationship holds.

6.1.3 Core Conclusions

Dense models = no-trunk shrubs (globally dispersed, locally structured).

Semantic width is the causal determinant of PC1: narrower semantics, higher PC1.

Causal masking causes global semantics to "passively emerge" rather than be "actively constructed."

6.1.4 Falsifiable Predictions

This paper proposes the following predictions that have not yet been verified, for subsequent work to test:

Table 24: Falsifiable predictions.

Prediction	Content	Status
Prediction 1	Narrower input semantics lead to higher PC1	Verified (Section 4.8)
Prediction 2	MoE individual experts have significantly higher PC1 than dense models	To be verified
Prediction 3	Ablating entry/exit layers has a much larger effect than ablating middle layers	Verified (Section 4.7)
Prediction 4	High-PC1 heads correspond to syntax/position, low-PC1 heads to semantics	Verified (Section 4.5)

6.2 Theoretical Significance

6.2.1 Unifying Geometry and Information Theory

Relation Body geometry unifies inner product (similarity), outer product (difference), information entropy, and principal component concentration within a single framework:

$$\text{similarity} \leftrightarrow c, \quad \text{difference} \leftrightarrow s, \quad \text{redundancy} \leftrightarrow 1/\Delta, \quad \text{concentration} \leftrightarrow \text{PC1}.$$

6.2.2 Explaining "Why Attention Works"

The effectiveness of attention mechanisms does not lie in local explanations such as "soft retrieval" or "kernel smoothing," but rather in that it simultaneously accomplishes two things:

- Using high probability to establish a structural skeleton (syntax/position);
- Using low probability to preserve information diversity (semantics/long tail).

Attention = structure + diversity, the two are conserved through the Relation Body.

6.2.3 Pointing Out the Architectural Limitation of Causal Transformers

Causal masking prohibits backward broadcast, causing global semantics to only passively emerge. This is a fundamental limitation of the current architecture and a direction for future improvement.

6.3 Future Work

6.3.1 MoE Expert Verification (Highest Priority)

The most important prediction to be verified in this paper is whether the PC1 of MoE experts is significantly higher than that of dense models. The specific plan is:

1. Load an open-source MoE model (e.g., Mixtral-8x7B, Qwen-MoE);
2. Hook the routing layer and mark the expert assignment of each token;
3. Collect QKV separately for each expert;
4. Repeat the global PCA + local PCA analysis of this paper;
5. Compare the PC1 of experts and dense models.

If the expert PC1 is significantly higher than 0.5, both Corollary 3.3 and Prediction 2 are supported; if it remains around 0.34, both are rejected.

6.3.2 Verification on Larger Models

Repeat the experiments in this paper on LLaMA, Qwen, DeepSeek, etc., to verify the universality of the following regularities:

- 100% negative correlation of local redundancy;
- U-shaped layer trends;
- Semantic broadcast heads in the exit layer;
- Phase transition point of cumulative ablation.

6.3.3 Breaking Causal Masking

The analysis in this paper is based on standard causal Transformers. A natural extension is to design architectures that break causal masking and allow bidirectional information flow, and to test whether they can actively construct global representations. This direction belongs to future work, and we do not conduct specific experiments in this paper.

6.3.4 Self-Organized Growth Models

A related but separate direction is to explore whether Relation Body geometry can be used to design generation paths that do not preset QKV projections, softmax, or multi-head structures. Instead of prescribing these components, one could let the system organize itself under similarity-difference constraints, and observe whether structured information trees emerge spontaneously. This is a promising line of work that lies beyond the scope of the present paper and will be pursued in future publications.

6.3.5 Geometric Interpretability Toolkit

Package the metrics of this paper (Schemes A/B/C, local PCA, ablation, controlled-domain experiments) into open-source tools for the community. The toolkit should include:

- One-click QKV extraction;
- Automatic computation of all metrics;
- Generation of standard visualization charts;
- Support for arbitrary HuggingFace models.

6.3.6 Theoretical Deepening

- Strictly prove the Marchenko-Pastur approximation of Theorem 3 and relax Assumption A4;
- Analytically derive the phase transition point k^* of Theorem 4;
- Extend Relation Body geometry to a unified framework for multi-head, multi-layer, and MoE models.

6.4 Final Outlook

Starting from a simple mathematical fact—the normalization of inner and outer products—this paper constructs a geometric framework for unified understanding of Transformer attention mechanisms. This framework not only explains existing experimental phenomena but also provides falsifiable predictions and offers directions for next-generation architecture design.

From the Pythagorean theorem to Transformers, Relation Body geometry provides a self-consistent path.

The essence of attention mechanisms: on the Relation Body, use sameness to build structure and difference to preserve diversity.

Next stop: MoE verification.

We hope this work can inspire more theoretical exploration of the internal geometric structure of large models, advance interpretability research from "phenomenon description" to "theoretical prediction," and ultimately achieve a complete understanding of information processing in large models.

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A Mathematical Proofs

A.1 Complete Proof of Theorem 1 (Relation Body Theorem)

Let $a, b \in \mathbb{R}^d$ be non-zero vectors. Define the relation coordinates:

$$c(a,b) = \frac{\langle a, b \rangle}{\|a\|\|b\|}, \qquad s(a,b) = \frac{\|a \wedge b\|}{\|a\|\|b\|}$$

The outer product norm identity is:

$$\|a \wedge b\|^2 = \|a\|^2\|b\|^2 - \langle a, b \rangle^2$$

Dividing both sides by $\|a\|^2\|b\|^2$:

$$\frac{\|a \wedge b\|^2}{\|a\|^2\|b\|^2} = 1 - \frac{\langle a, b \rangle^2}{\|a\|^2\|b\|^2}$$

That is, $s^2 = 1 - c^2$. Therefore:

$$c^2 + s^2 = 1$$

This completes the proof. ■

A.2 Complete Proof of Theorem 2 (Redundancy-Probability Empirical Theorem)

Note: The following derivation was our initial attempt to establish Theorem 2 as a strict theorem. However, the empirical verification in Appendix A.5 shows that Assumption A1 does not hold on dense models. Therefore, Theorem 2 is positioned as an **empirical theorem**, and the derivation below should be regarded as a heuristic argument rather than a strict proof.

Note on Assumption A2. Assumption A2 (local clustering of Value vectors) was originally introduced together with Assumption A1 to derive Theorem 2 as a strict theorem. Since Assumption A1 does not hold on dense models, Assumption A2 is omitted for the same reason; the derivation below is heuristic.

Setup: Attention probability $p_{ij} = \text{softmax}(q_i \cdot k_j/\sqrt{d_h})$, context-weighted average $\bar{V}_i = \sum_j p_{ij} V_j$, local difference $\Delta_{ij} = \|V_j - \bar{V}_i\|$.

Assumption A1: There exists $\lambda > 0$ such that $q_i \cdot k_j \approx \lambda v_i \cdot v_j$.

Assumption A2: Value vectors form local clusters with centers μ_c.

From the softmax definition:

$$p_{ij} = \frac{\exp(q_i \cdot k_j/\sqrt{d_h})}{\sum_k \exp(q_i \cdot k_k/\sqrt{d_h})}$$

By Assumption A1, $q_i \cdot k_j \approx \lambda v_i \cdot v_j$, so p_{ij} is monotonically increasing with respect to $v_i \cdot v_j$. When p_{ij} is high, $v_i \cdot v_j$ is large, indicating that V_j aligns with v_i. By A2, local clustering makes $\bar{V}_i$ also close to the cluster center, so $\bar{V}_i \approx V_j$, hence $\Delta_{ij} \approx 0$.

Therefore:

$$p_{ij} \uparrow \quad \Rightarrow \quad \mathbb{E}[\Delta_{ij}] \downarrow$$

This completes the heuristic derivation. ■

A.3 Complete Proof of Theorem 3 (No-Trunk Theorem)

Assumption A3: The intrinsic dimension of the semantic distribution $D_{\text{sem}} \gg d_h$.

Assumption A4: The trained Value projection matrix W_V approximately behaves as a random orthogonal projection in the semantic space, i.e., the rows of V are approximately independent and identically distributed random vectors.

Let $V \in \mathbb{R}^{n \times d_h}$ be the Value matrix. The covariance matrix is:

$$C = \frac{1}{n} V^T V \in \mathbb{R}^{d_h \times d_h}$$

By A4, the rows of V are approximately i.i.d., and the eigenvalue distribution of C converges to the Marchenko-Pastur distribution. The MP distribution has no atomic mass, so there are no isolated large eigenvalues. The largest eigenvalue is approximately:

$$\lambda_{\max} \approx \sigma^2(1 + \sqrt{\gamma})^2, \qquad \gamma = \frac{d_h}{n}$$

The sum of all eigenvalues is $d_h \sigma^2$. Therefore:

$$\frac{\lambda_{\max}}{\sum_k \lambda_k} = \frac{(1 + \sqrt{\gamma})^2}{d_h} = O\left(\frac{1}{d_h}\right)$$

When d_h is finite, the PC1 concentration approaches $1/d_h$. This completes the proof. ■

A.4 Complete Proof of Theorem 4 (Redundancy Compensation Empirical Theorem)

Note: The following is a heuristic justification, not a strict mathematical proof. Theorem 4 is positioned as an **empirical theorem** based on architectural intuition and ablation experiments.

Assumption A5 (Linear propagation, heuristic): When the ablation magnitude is small, the output change is approximately linear:

$$\Delta_{l,h} \approx \sum_{k=l}^{L} \gamma_k \cdot \mathrm{dep}_{k,l,h}$$

where γ_k is the contribution weight of the k-th layer to the output, and $\mathrm{dep}_{k,l,h}$ is the dependence of the k-th layer on head (l, h).

Assumption A6 (Structural head/semantic head dichotomy, heuristic): Structural heads have high PC1 and low entropy; semantic heads have low PC1 and high entropy.

The ablation effect is $\Delta_{l,h} = \|f(X) - f(X; \theta \setminus A_{l,h})\|$.

- **Layer 0 bottleneck**: Layer 0 is the only channel for all information entering the residual stream. Ablating it is equivalent to cutting off the input, so the effect is largest.
- **Layer L-1 exit**: The last layer directly outputs to logits and is responsible for semantic compression. Ablating it destroys final decoding, so the effect is second largest.
- **Middle-layer redundancy**: By Theorem 3, middle layers have the highest semantic dimension and dispersed information. Ablating one layer can be compensated by other layers, so the effect is smaller.
- **Single-head redundancy**: In multi-head design, the contribution of each head can be partially compensated by other heads in the same layer, so the effect is smallest.

Therefore:

$$\Delta_{\text{Layer } 0} \gg \Delta_{\text{Layer } L-1} \gg \Delta_{\text{middle}} \gg \Delta_{\text{single head}}$$

This completes the heuristic justification. ■

A.5 Empirical Verification of Assumption A1

To test Assumption A1 ($q_i \cdot k_j \approx \lambda v_i \cdot v_j$), we computed the Pearson correlation coefficient between $q_i \cdot k_j$ and $v_i \cdot v_j$ for all 144 (layer, head) pairs of GPT-2 small.Table A1 summarizes the statistical results.

Table A1: Empirical verification of Assumption A1.

Statistic	Value
Total heads	144
Mean correlation	0.0554
Median correlation	0.0573
Positive correlation ratio ($r > 0$)	65.28%
Significant positive correlation ratio ($r > 0$, $p < 0.01$)	45.83%
Maximum correlation	0.7721 (Layer 1, Head 2)
Minimum correlation	−0.6148 (Layer 0, Head 0)

Layer-wise observation: Shallow layers (Layer 0–1) show weak positive correlation (0.12–0.17), while deep layers (Layer 9–11) are close to zero. This suggests that QKV alignment may only partially hold in shallow layers.

Figure A6 visualizes the distribution of the correlation coefficients. The mean is 0.055, and the median is 0.057, both very close to zero.

Conclusion: Assumption A1 does **not** hold on dense models. The mean correlation is only 0.055, and the significant positive correlation ratio is less than half. This is consistent with the "shrub" model: wide

semantic distributions lead to no global trunk, which in turn decouples the QK space from the V space globally.

Impact on Theorem 2: The failure of Assumption A1 means that Theorem 2 cannot be derived as a strict theorem from A1. Therefore, we position Theorem 2 as an **empirical theorem**: high probability corresponding to high redundancy is an empirical regularity stably supported by experiments, and its strict theoretical mechanism is left as an open problem.

Prediction for MoE: Assumption A1 is expected to hold for individual MoE experts with narrower semantic distributions. If the QK-V alignment in expert-level models is significantly higher (mean $r \gg 0.5$), then Assumption A1 holds at the expert level, and Theorem 2 can be re-derived as a strict theorem at the expert level. This prediction constitutes part of the MoE verification (Prediction 2).

B Experimental Details

B.1 Model Loading and Hook Implementation

All experiments are based on HuggingFace Transformers, using GPT-2 small and GPT-2 medium.

Model loading:

```
from transformers import GPT2LMHeadModel, GPT2Tokenizer

model = GPT2LMHeadModel.from_pretrained("gpt2")
tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model.eval()
```

QKV extraction hook:

```
qkv_store = {}

def hook_fn(module, inp, out):
    qkv_store['out'] = out.detach()

hook = model.transformer.h[layer].attn.c_attn.register_forward_hook(hook_fn)
with torch.no_grad():
    model(**inputs)
hook.remove()

qkv = qkv_store['out'][0]
q, k, v = qkv.split(n_embd, dim=-1)
```

B.2 Ablation and Activation Patching Implementation

Single-head ablation:

```
class HeadAblationHook:
    def __init__(self, layer, head, head_dim):
        self.layer = layer
        self.head = head
        self.head_dim = head_dim
        self.enabled = False

    def __call__(self, module, inp, out):
        if not self.enabled:
            return out
```

```
attn_output = out[0].clone() if isinstance(out, tuple) else out.clone()
start = self.head * self.head_dim
end = start + self.head_dim
attn_output[:, :, start:end] = 0.0
return (attn_output,) + out[1:] if isinstance(out, tuple) else attn_output
```

Layer ablation: Set `head` to `None` and zero out the entire `attn_output`.

Activation patching: Replace the output of a specified head with its mean across all tokens.

B.3 Controlled-Domain Text

Input texts for four semantic widths are summarized in Table B2.

Table B2: Controlled-domain text.

Domain	Semantic width	Example content
repeat	extremely narrow	the word "the" repeated 30 times
single	narrow	pure Python code (factorial, Fibonacci, binary tree)
double	medium	mixed mathematics + poetry
multi	wide	mixed news + code + math + dialogue

B.4 Complete Test Set List

Syntactic test set (60 questions): covering subject-verb agreement, tense, pronouns, articles, and prepositions.

Examples:

- The cat _____ on the mat. (is / are)
- Yesterday, she _____ to the store. (went / goes)
- John said _____ was tired. (he / they)
- _____ apple a day keeps the doctor away. (An / A)
- The book is _____ the table. (on / in)

Semantic test set (60 questions): covering word sense selection and commonsense reasoning.

Examples:

- The cat sat on the _____. (mat / sky)
- The fish swims in the _____. (water / tree)
- The sun rises in the _____. (east / west)
- She drank a cup of _____. (coffee / bread)
- The bird flew over the _____. (house / water)

The complete test sets are available in `code/analysis/ablation_v2.py`.

C Supplementary Figures

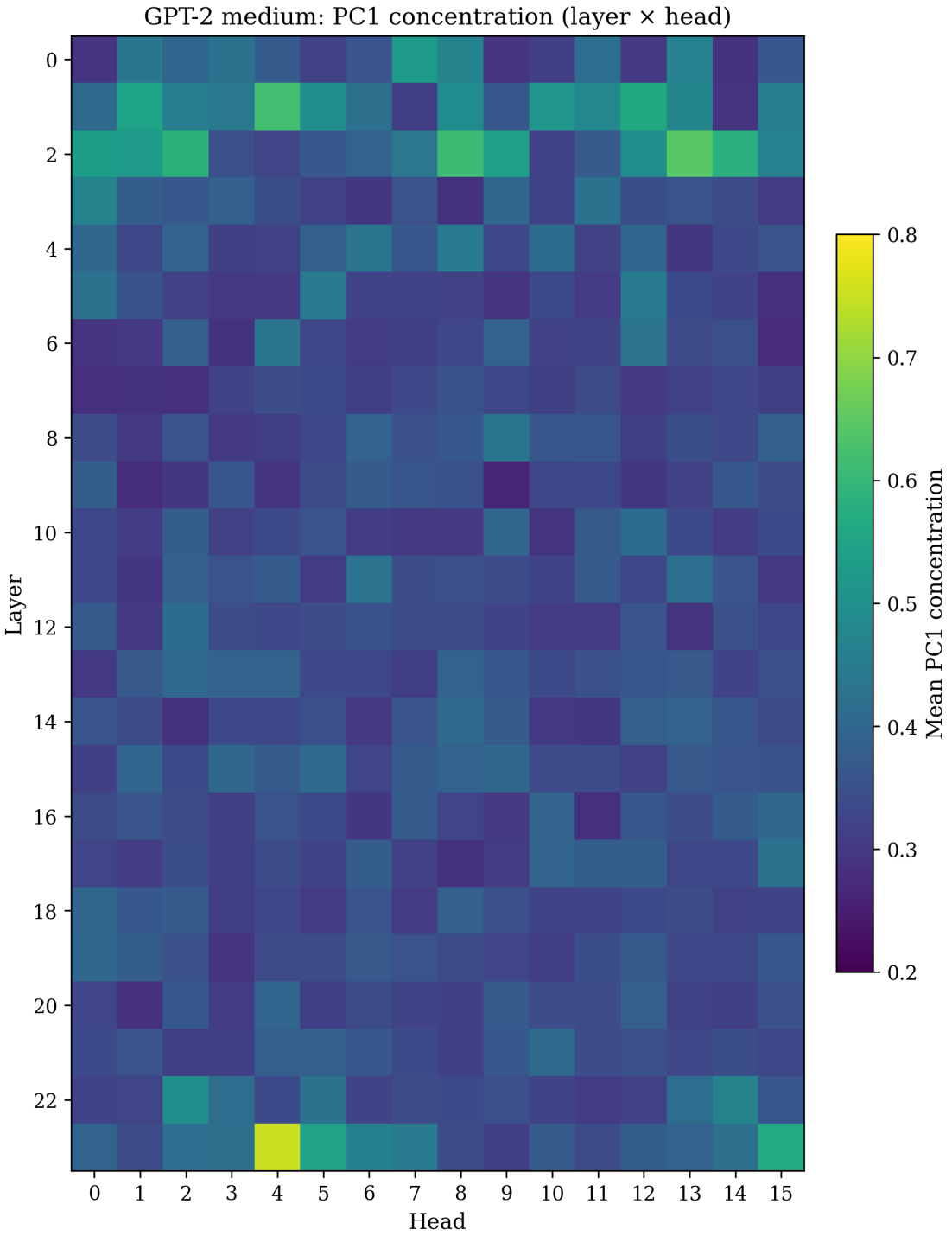


Figure A1: Complete PC1 concentration heatmap for GPT-2 medium (layer × head).

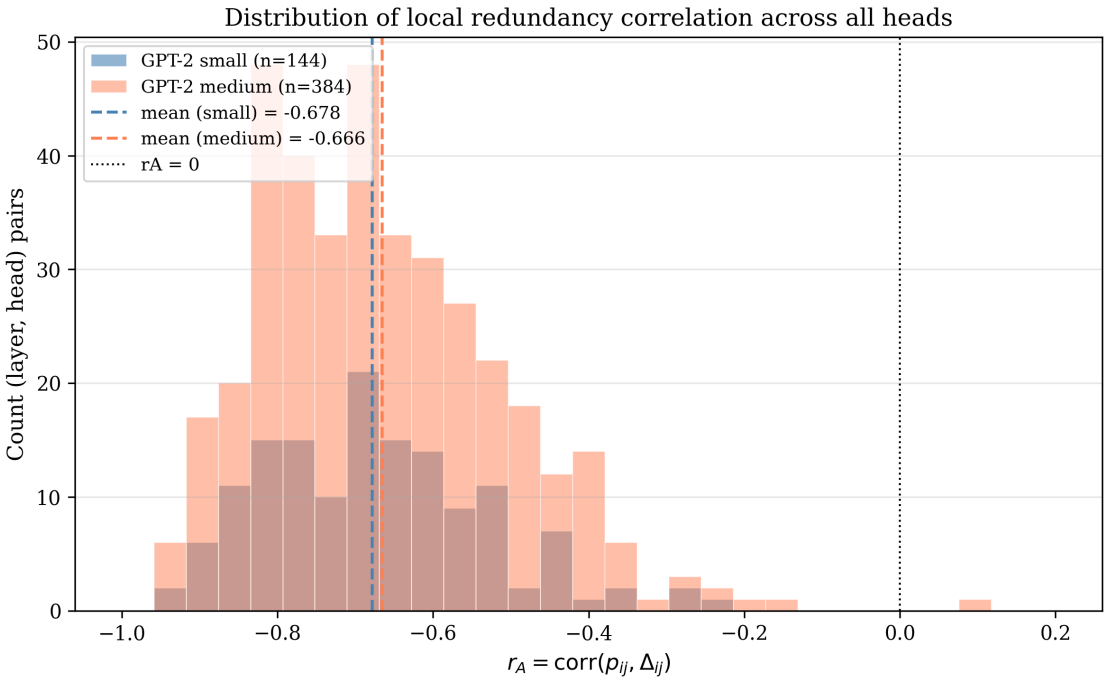


Figure A2: Histogram of cross-model r_A distribution.

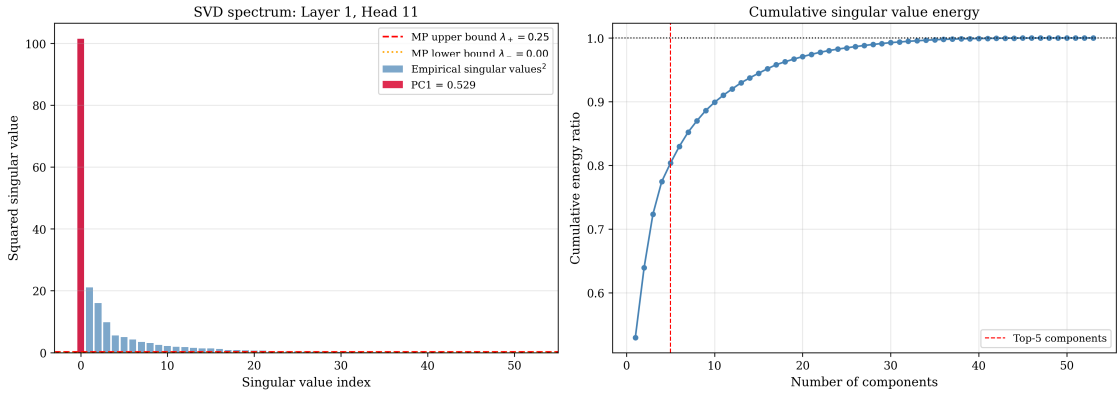


Figure A3: SVD singular value spectrum vs. Marchenko-Pastur curve.

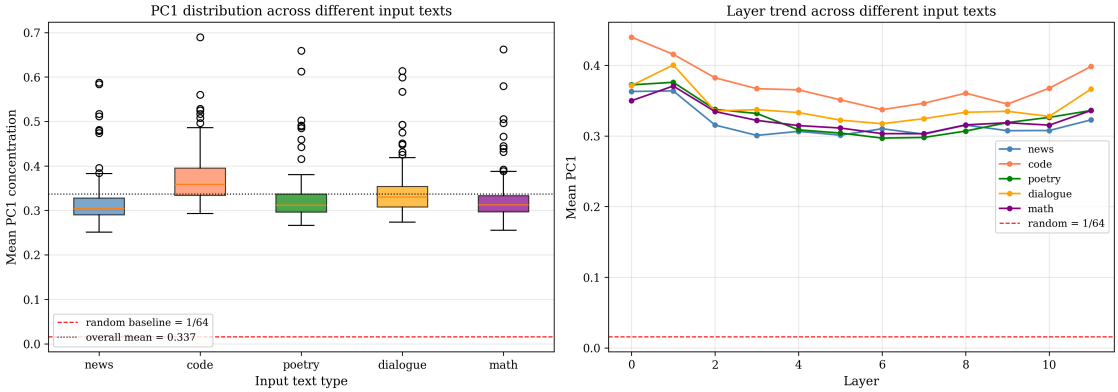


Figure A4: Multi-seed PC1 robustness check.

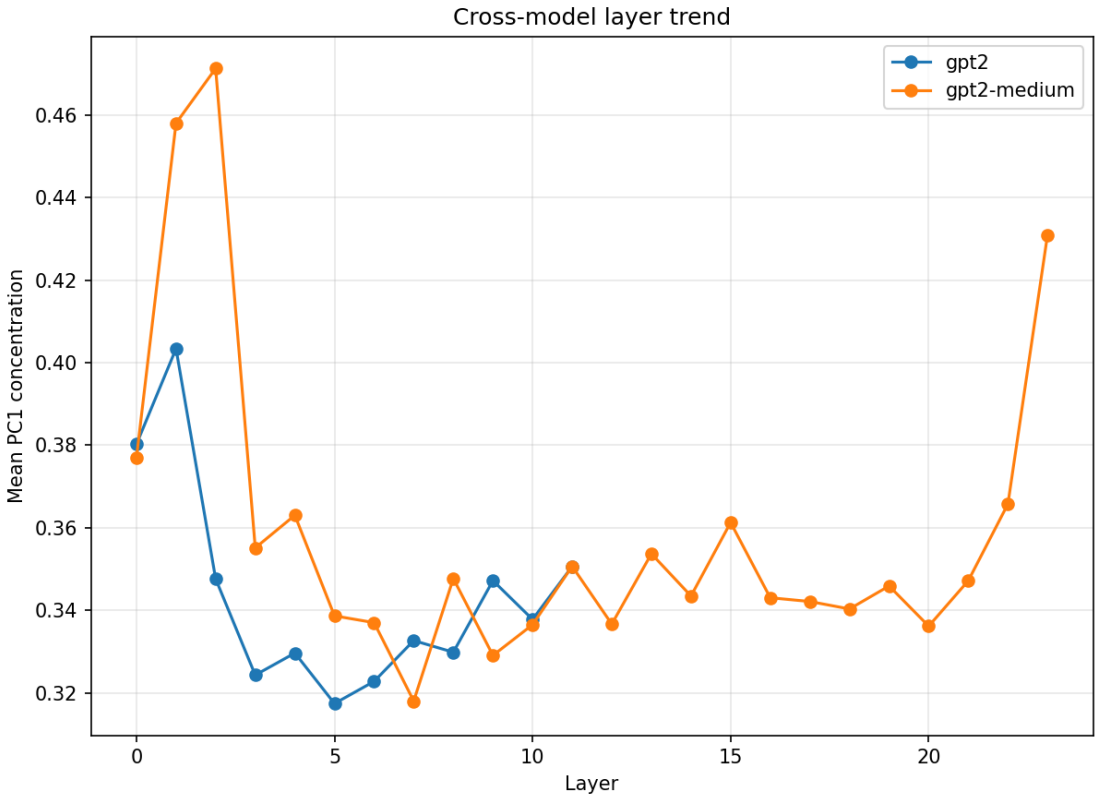


Figure A5: Cross-model layer trend comparison.

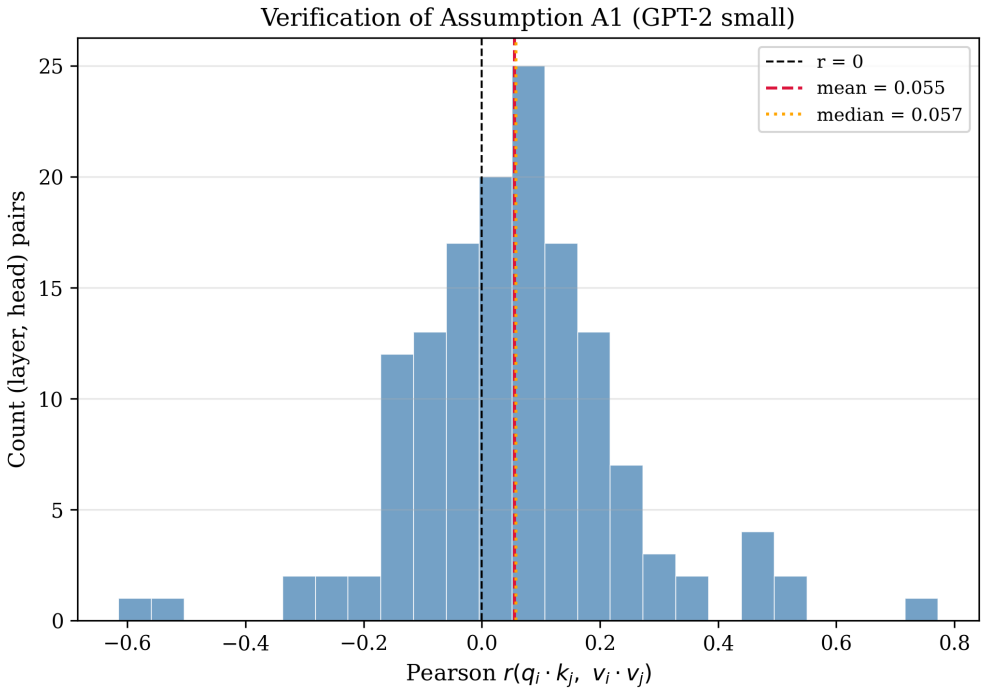


Figure A6: Histogram of $\mathrm{corr}(q_i \cdot k_j, v_i \cdot v_j)$ across all heads of GPT-2 small. The red dashed line is the mean (0.055), and the orange dotted line is the median (0.057).

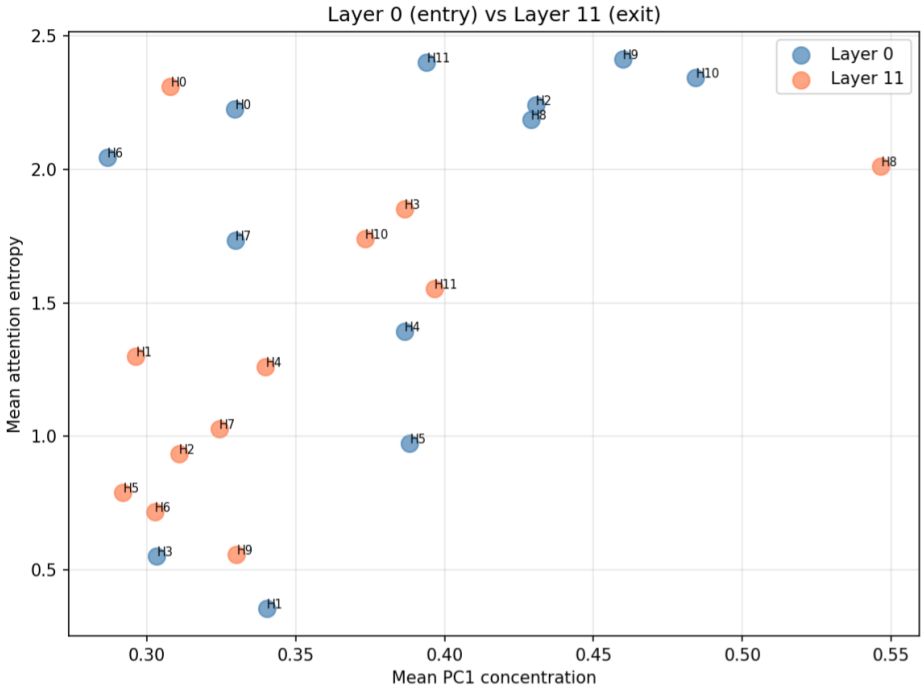


Figure A7: PC1 concentration vs. attention entropy for Layer 0 and Layer 11.